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CANDIDATE NUMBER

**2020** Trial Examination

# Form VI Mathematics Extension 1

Monday 17th August 2020

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## General Instructions

- Reading time — 10 minutes
- Working time — 2 hours
- Attempt all questions.
- Write using black pen.
- Calculators approved by NESA may be used.
- A loose reference sheet is provided separate to this paper.

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**Total Marks: 70**

### Section I (10 marks) Questions 1–10

- This section is multiple-choice. Each question is worth 1 mark.
- Record your answers on the provided answer sheet.

### Section II (60 marks) Questions 11–14

- Relevant mathematical reasoning and calculations are required.
- Start each question in a new booklet.

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## Collection

- If you use multiple booklets for a question, place them inside the first booklet for the question.
- Arrange your solutions in order.
- Write your candidate number on this page, on the start of the separate section and on the multiple choice sheet.
- Place everything inside this question booklet.

## Checklist

- Reference sheet
- Multiple-choice answer sheet
- 4 booklets per boy
- Candidature: 128 pupils

**Writer: RCF**

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## Section I

Questions in this section are multiple-choice.

Choose a single best answer for each question and record it on the provided answer sheet.

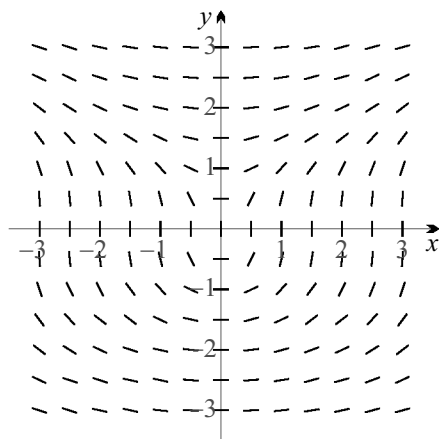
1. A projectile has an initial velocity vector  $\mathbf{v} = \begin{bmatrix} 2\sqrt{3} \\ 2 \end{bmatrix}$ . Which of the following is the correct statement of its initial speed and angle of projection from the horizontal?

(A) 4 m/s at  $30^\circ$   
 (B)  $\sqrt{10}$  m/s at  $30^\circ$   
 (C) 4 m/s at  $60^\circ$   
 (D)  $\sqrt{10}$  m/s at  $60^\circ$

2. What is the coefficient of the  $x^3$  term in the expansion of  $(2 - 3x)^8$ ?

(A) 56  
 (B) -1512  
 (C) 1512  
 (D) -48 384

3.



Which of the following differential equations could produce the slope field shown above?

(A)  $\frac{dy}{dx} = \frac{x}{y^2}$   
 (B)  $\frac{dy}{dx} = xy$   
 (C)  $\frac{dy}{dx} = \frac{x}{y}$   
 (D)  $\frac{dy}{dx} = x^2y$

4. Given  $\underline{a} = \begin{bmatrix} 5 \\ 5 \end{bmatrix}$  and  $\underline{b} = \begin{bmatrix} 4 \\ -2 \end{bmatrix}$ , which of the following represents  $\text{proj}_{\underline{b}} \underline{a}$ ?
- (A)  $\begin{bmatrix} 40 \\ -20 \end{bmatrix}$
- (B)  $\begin{bmatrix} 20 \\ -10 \end{bmatrix}$
- (C)  $\begin{bmatrix} 2 \\ -1 \end{bmatrix}$
- (D)  $\begin{bmatrix} \frac{4}{5} \\ -\frac{2}{5} \end{bmatrix}$
5. Which expression is equivalent to  $1 - \cos 2x$ ?
- (A)  $\sin^2 x - \cos^2 x$
- (B)  $2(\sin^2 x + 1)$
- (C)  $2\sin^2 x$
- (D)  $2(1 + \cos^2 x)$
6. A school mathematics department consists of 5 male and 5 female teachers. How many different exam-writing committees comprising three teachers could be formed if there must be at least one teacher of each sex on the committee?
- (A) 50
- (B) 100
- (C) 120
- (D) 600
7. Which of the following is a correct primitive of  $\tan x$ ?
- (A)  $\sec^2 x + c$
- (B)  $\ln |A \sin x|$
- (C)  $\frac{1}{1+x^2} + c$
- (D)  $-\ln |A \cos x|$
8. Which of the following is equivalent to  $\int_0^{\frac{\pi}{3}} \cos^5 x \sin x \, dx$ , after applying the substitution  $u = \cos x$ ?
- (A)  $\int_0^{\frac{\pi}{3}} u^5 \, du$
- (B)  $-\int_{\frac{1}{2}}^1 u^5 \, du$
- (C)  $-\int_0^{\frac{\pi}{3}} u^5 \, du$
- (D)  $\int_{\frac{1}{2}}^1 u^5 \, du$

9. What is the value of  $\cos^{-1}(\sin \alpha)$ , where  $\frac{\pi}{2} < \alpha < \pi$ ?

(A)  $\pi - \alpha$

(B)  $\frac{\pi}{2} - \alpha$

(C)  $\alpha - \frac{\pi}{2}$

(D)  $\alpha + \frac{\pi}{2}$

10. In trying to solve  $\frac{x-2}{x-3} < \frac{4}{\sqrt{x-2}}$  over its natural domain in the set of real numbers, three students produce the following inequalities.

Student I  $(x-2)\sqrt{x-2} < 4(x-3)$

Student II  $\sqrt{(x-2)^3}(x-3) < 4(x-3)^2$

Student III  $(x-2)^2(x-3) < 4(x-3)^2\sqrt{x-2}$

Which students are still on track to obtain the correct solution set?

(A) Just student II

(B) Student III only

(C) Both students II and III but not student I

(D) All three students

**End of Section I**

**The paper continues in the next section**

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CANDIDATE NUMBER

## Section II

This section consists of long-answer questions.

Marks may be awarded for reasoning and calculations.

Marks may be lost for poor setting out or poor logic.

Start each question in a new booklet.

**QUESTION ELEVEN** (15 marks) Start a new answer booklet. **Marks**

- (a) The polynomial  $P(x) = x^3 - x^2 + kx - 4$  has a factor  $(x - 1)$ . Find the value of  $k$ . 1
- (b) Find the exact value of  $\int_0^2 \frac{1}{4 + x^2} dx$ . 2
- (c) Consider the curve  $f(x) = 3 \sin^{-1} \left( \frac{x}{2} \right)$ .
- (i) Sketch the curve, clearly indicating the coordinates of any intercepts with the axes and any endpoints. 2
- (ii) Find the exact gradient of the tangent to the curve at the point where  $x = \frac{1}{2}$ . 2
- (d) Use the substitution  $u = 2x - 1$  to find  $\int 4x\sqrt{2x - 1} dx$ . 2
- (e) The equation  $x^3 + bx^2 + 6x + d = 0$  has roots  $1 + \sqrt{3}$ ,  $1 - \sqrt{3}$  and 4. Use the sum and product of roots to find the integers  $b$  and  $d$ . 2
- (f) A particle moves in two dimensional space where  $\underline{i}$  and  $\underline{j}$  are unit vectors in the  $x$  and  $y$  directions respectively. At time  $t$  seconds its displacement from the origin is given by  $\underline{r} = (6t - 4t^2)\underline{i} + 2t\underline{j}$  where all lengths are measured in metres.
- (i) Write down the particle's velocity vector in component form. 1
- (ii) Find the speed of the particle when  $t = 2$ . 1
- (iii) Show that the equation of the path of the particle is  $x = 3y - y^2$ . 2

**QUESTION TWELVE** (15 marks) Start a new answer booklet.**Marks**

(a) Solve the equation  $\sin 2x = \sqrt{2} \sin x$ , for  $0 \leq x \leq 2\pi$ . 3

(b) Solve the initial value problem where  $\frac{dy}{dx} = 2xe^{-y}$  given  $y(0) = \ln 4$ . Express your answer with  $y$  as the subject. 3

(c) Points  $P(4, -6)$ ,  $Q(1, 2)$ ,  $R(7, 5)$  form a triangle  $PQR$  in the Cartesian Plane.

(i) Find the vectors  $\overrightarrow{QP}$  and  $\overrightarrow{QR}$ , representing two sides of this triangle. Give your answer in component form. 1

(ii) Use the dot product to find angle  $PQR$ . Give your answer correct to the nearest degree. 2

(d) Use mathematical induction to prove that, for positive integers  $n$ : 3

$$1 + 3 + 9 + \cdots + 3^{n-1} = \frac{1}{2}(3^n - 1).$$

(e) Write  $\cos^2 2x$  in terms of  $\cos 4x$  and hence evaluate  $\int_0^{\frac{\pi}{6}} \cos^2 2x \, dx$ . 3

**QUESTION THIRTEEN** (15 marks) Start a new answer booklet.

Marks

(a) (i) Express  $\sqrt{3}\cos\theta - \sin\theta$  in the form  $R\cos(\theta + \alpha)$ , where  $R > 0$  and  $0 < \alpha < \frac{\pi}{2}$ . 2(ii) Hence, or otherwise, solve  $\sqrt{3}\cos\theta - \sin\theta = -1$ , for  $0 \leq \theta \leq 2\pi$ . 2(b) The rate at which a cool object warms in air is proportional to the difference between its temperature  $T$ , in degrees Celsius, and the constant ambient temperature  $A$  °C of the surrounding air. This rate can be expressed by the differential equation:

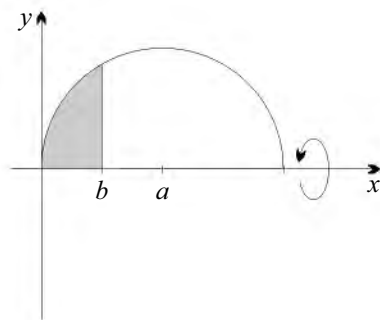
$$\frac{dT}{dt} = k(A - T)$$

where  $t$  is time in minutes and  $k$  is a positive constant. The solution of this differential equation is  $T = A + Be^{-kt}$ , where  $B$  is a constant. (You need NOT show this.)

A bottle of baby milk is at 4 °C when it is removed from a refrigerator and placed on the kitchen bench where the room temperature is 22 °C. Five minutes later it has warmed to 12 °C. 3

Find the temperature of the milk after a further three minutes sitting on the bench. Give your answer correct to the nearest degree.

(c)

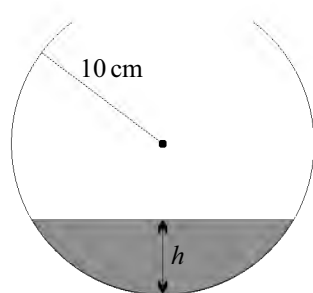


Consider the region enclosed by the upper semicircle  $y = \sqrt{a^2 - (x - a)^2}$  and the vertical line  $x = b$  where  $0 < b < 2a$ , shown shaded in the diagram above.

(i) A spherical cap is generated by rotating this region around the  $x$ -axis. Show that the volume  $V$ , in cubic units, of this solid is given by: 3

$$V = \frac{\pi b^2}{3}(3a - b).$$

(ii)



In the diagram above, a spherical vase of radius 10 cm is being filled with water at a constant rate  $90 \text{ cm}^3/\text{min}$ .

Let  $h$  cm be the depth of the water after  $t$  minutes. Find the rate at which the depth of the water is rising at the instant when the depth is 5 cm. Give your answer in terms of  $\pi$ . 2

(d) Use  $t$  formulae to solve  $\sin x - 7\cos x = 5$ , for  $0 \leq x \leq 2\pi$ . 3

**QUESTION FOURTEEN** (15 marks) Start a new answer booklet.**Marks**

- (a) Let the function  $f(x) = \sec x$  be defined for the restricted domain  $0 < x < \frac{\pi}{2}$ , so that its inverse  $f^{-1}(x)$  is also a function.

(i) Show that  $f^{-1}(x) = \cos^{-1}\left(\frac{1}{x}\right)$  and clearly state the domain of  $f^{-1}(x)$ . 2

(ii) Hence find  $\frac{d}{dx}\left(f^{-1}(x)\right)$  and describe the behaviour of the graph of  $y = f^{-1}(x)$  as  $x \rightarrow \infty$ . 2

- (b) Data suggests that the number of cases of infection from a particular disease tends to fluctuate between two values over a period of approximately six months.

Let  $P$  be the number of cases after  $t$  months, where  $P$  is measured in thousands. Initially there are 1000 cases.

- (i) Suppose that  $P$  is modelled by the equation: 2

$$P = \frac{2}{2 - \sin t}.$$

Verify that  $P$  satisfies the differential equation  $\frac{dP}{dt} = \frac{1}{2}P^2 \cos t$ .

- (ii) An alternative model is proposed with a different differential equation: 4

$$\frac{dP}{dt} = \frac{1}{2}(2P^2 - P) \cos t.$$

Use the result that  $\frac{1}{P(2P-1)} = \frac{2}{2P-1} - \frac{1}{P}$  to solve this differential equation, showing that:

$$P = \frac{1}{2 - e^{\frac{1}{2} \sin t}}$$

and that this new solution also satisfies the initial condition.

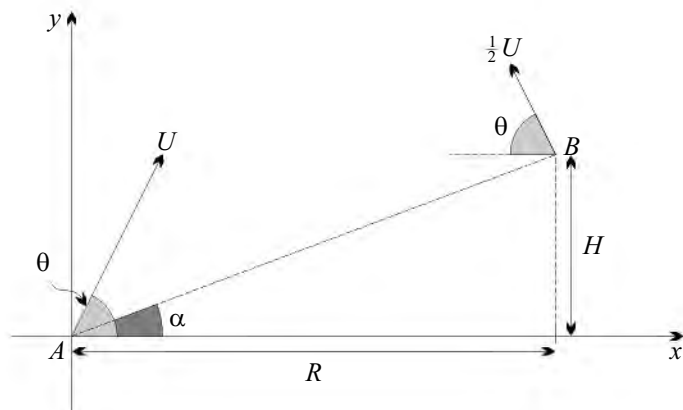
- (iii) Find the greatest and least values of  $P$  predicted by both models for  $t \geq 0$ . Give your answers in exact form and then rounded to three decimal places to enable easy comparison between the two models. 2

**The question continues on the next page**

**QUESTION FOURTEEN** (Continued)

(c)

3



In the diagram above, points  $A$  and  $B$  are separated by a horizontal distance  $R$  and point  $B$  is located  $H$  metres higher than  $A$ . Define the angle of inclination of  $B$  from  $A$  as  $\alpha$ . Define the origin for both motions at point  $A$  and positive directions as right and up respectively.

At the same instant, identical projectiles are launched from each location directed towards each other. The projectile from  $A$  is fired with initial speed  $U$  m/s at an angle  $\theta$  above the horizontal while the object launched from  $B$  has only half as much initial speed but the same angle of elevation above the horizontal.

Consequently, the equations of motion for the projectile from  $A$ ,  $t$  seconds after launch, are as follows:

$$x_A = Ut \cos \theta \quad y_A = Ut \sin \theta - \frac{1}{2}gt^2.$$

Similarly for the projectile from  $B$ :

$$x_B = R - \frac{Ut \cos \theta}{2} \quad y_B = H + \frac{Ut \sin \theta}{2} - \frac{1}{2}gt^2.$$

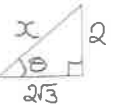
[Do NOT prove these equations.]

Given that the projectiles collide, show that this requires

$$\tan \alpha = \frac{1}{3} \tan \theta.$$

————— **END OF PAPER** —————

Extension 1, 2020 Trial.

①  $\begin{pmatrix} 2\sqrt{3} \\ 2 \end{pmatrix}$    $x^2 = 2^2 + (2\sqrt{3})^2 = 4 + 12 = 16$   
 $x = 4$   
 $\tan \theta = \frac{2}{2\sqrt{3}} = \frac{1}{\sqrt{3}}$   
 $\theta = 30^\circ$  hence (A)

②  $(2-3x)^8 = \dots + {}^8C_5 \cdot 2^5 \cdot (-3x)^3 + \dots$   
 $= \dots + 56 \times 32 \times (-27)x^3 + \dots$  hence (D)  
 $= \dots - 48384x^3 + \dots$

③  $x=0$   $\frac{dy}{dx} = 0$   $y=0$  Gradients vertical hence not (B) or (D)  
 1<sup>st</sup> quad  $\frac{dy}{dx} > 0$  2<sup>nd</sup> quad  $\frac{dy}{dx} < 0$  3<sup>rd</sup> quad  $\frac{dy}{dx} < 0$  hence not (C)  
 hence (A)

④  $\text{proj}_{\vec{b}} \vec{a} = \frac{\vec{a} \cdot \vec{b}}{\vec{b} \cdot \vec{b}} \vec{b}$   
 $= \frac{20 \cdot 10}{4^2 + 2^2} \begin{bmatrix} 4 \\ 2 \end{bmatrix}$   
 $= \frac{10}{20} \begin{bmatrix} 4 \\ 2 \end{bmatrix}$   
 $= \begin{bmatrix} 2 \\ 1 \end{bmatrix}$  hence (C)

⑤  $1 - \cos 2x = 1 - (\cos^2 x - \sin^2 x)$   
 $= \sin^2 x + 1 - \cos^2 x$  not (A)  
 $= \sin^2 x + 1 - (1 - \sin^2 x)$   
 $= 2\sin^2 x$  hence (C)

⑥ 2M1F  ${}^5C_2 \times {}^5C_1$   
 2F1M  ${}^5C_2 \times {}^5C_1$   
 At least one of each sex  $2 \times {}^5C_2 \times {}^5C_1 = 2 \times \frac{5 \times 4}{2} \times 5 = 100$  hence (B)

⑦  $\int \tan x dx = \int \frac{-\sin x}{\cos x} dx$   
 $= -\ln |\cos x| + C$  hence (D) not (B)  
 $= -\ln |\sec x|$  if  $C = -\ln A$

⑧  $\int_0^{\pi/3} \cos^5 x \sin x dx$   $u = \cos x$   
 $du = -\sin x dx$   
 $= \int_{u=1}^{\frac{1}{2}} u^5 (-du)$ 

|   |   |               |
|---|---|---------------|
| x | 0 | $\pi/3$       |
| u | 1 | $\frac{1}{2}$ |

  
 $= -\int_1^{\frac{1}{2}} u^5 du$   
 $= \int_{\frac{1}{2}}^1 u^5 du$  (Reversing limits) hence (D)

⑨  $\cos^{-1}(\sin \alpha)$   $\frac{\pi}{2} < \alpha < \pi$  is 2<sup>nd</sup> quad  
  
 $\sin \alpha > 0$   
 $\therefore \cos^{-1}(\sin \alpha) = \cos^{-1}(\cos(\frac{\pi}{2} - \alpha))$   
 $= \frac{\pi}{2} - \alpha$   
 $\therefore \angle ABC = \frac{\pi}{2} - \angle BAC$   
 $= \frac{\pi}{2} - (\pi - \alpha)$   
 $= \alpha - \frac{\pi}{2}$  hence (C)  
 alternatively: use test case  
 eg  $\cos^{-1}(\sin \frac{5\pi}{6}) = \cos^{-1}(\frac{1}{2}) = \frac{\pi}{3}$   
 If  $\alpha = \frac{5\pi}{6}$   $\pi - \alpha = \frac{\pi}{6}$  not (A)  
 $\frac{\pi}{2} - \alpha = -\frac{\pi}{6}$  not (B)  
 $\alpha + \frac{\pi}{2} = \frac{8\pi}{6}$  not (D)

⑩ Domain  $x \neq 3$  and  $x > 2$   
 $\frac{x(x-2)}{x(x-3)^2}$   
 Student I in error since sign of  $x-3$  is indeterminate  
 Student II is correct  
 Student III is also correct, multiplication by extra factor of  $\sqrt{x-2}$  on both sides is unnecessary but still gives a valid inequality hence (C)

QV11

$$a) P(x) = x^3 - x^2 + kx - 4$$

$$P(1) = 0 \quad 1 - 1 + k - 4 = 0$$

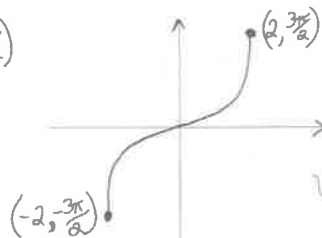
$$\underline{k = 4}$$

$$b) \int_0^2 \frac{1}{4+x^2} dx = \left[ \frac{1}{2} \tan^{-1} \frac{x}{2} \right]_0^2$$

$$= \frac{1}{2} (\tan^{-1} 1 - \tan^{-1} 0)$$

$$= \underline{\frac{\pi}{8}}$$

$$c) f(x) = 3 \sin^{-1} \left( \frac{x}{2} \right)$$



$$f'(x) = 3 \times \frac{1}{\sqrt{1 - \left(\frac{x}{2}\right)^2}} \times \frac{1}{2}$$

$$= \frac{3}{2\sqrt{1 - \frac{x^2}{4}}}$$

$$f'\left(\frac{1}{2}\right) = \frac{3}{2\sqrt{1 - \frac{1}{16}}}$$

$$= \frac{3}{2\sqrt{\frac{15}{16}}}$$

$$= \frac{6}{\sqrt{15}}$$

$$= \frac{6\sqrt{15}}{15}$$

$$= \underline{\frac{2\sqrt{15}}{5}}$$

$$d) \int 4x \sqrt{2x-1} dx$$

$$= \int (u+1) \sqrt{u} du$$

$$= \int u^{3/2} + u^{1/2} du$$

$$= \frac{2}{5} u^{5/2} + \frac{2}{3} u^{3/2} + C$$

$$= \underline{\frac{2}{5} (2x-1)^{5/2} + \frac{2}{3} (2x-1)^{3/2} + C}$$

$$\text{Let } u = 2x-1$$

$$du = 2dx$$

f)

$$r = (6t - 4t^2) \hat{i} + 2t \hat{j}$$

$$i) \underline{v} = \frac{dr}{dt} = (6 - 8t) \hat{i} + 2 \hat{j} \quad \text{m/s} \quad \checkmark$$

Do not penalize units

$$ii) \underline{v}_{t=2} = -10 \hat{i} + 2 \hat{j}$$

$$\text{speed} = |\underline{v}| = \sqrt{(-10)^2 + 2^2}$$

$$= \sqrt{104}$$

$$= \underline{2\sqrt{26} \text{ m/s}} \quad \checkmark$$

(Penalize units here)

$$iii) x = 6t - 4t^2 \quad y = 2t$$

$$\text{from } ② \quad t = \frac{y}{2}$$

$$\text{sub } t \text{ into } ① \quad x = 6\left(\frac{y}{2}\right) - 4\left(\frac{y}{2}\right)^2$$

$$x = 3y - y^2 \quad \checkmark$$

$$e) x^3 + bx^2 + 6x + d = 0$$

has roots  $1+\sqrt{3}, 1-\sqrt{3}$  and 4

$$\Sigma x = -\frac{b}{a} \therefore 6 = -b \Rightarrow b = -6$$

$$\Sigma x^2 = -\frac{d}{a} \therefore 4(1+\sqrt{3})(1-\sqrt{3}) = -d$$

$$-8 = -d$$

$$\underline{d = 8}$$

$$\text{Alternatively: } (x-4)(x-1-\sqrt{3})(x-1+\sqrt{3})$$

$$= (x-4)(x^2 - 2x - 2)$$

$$x^3 - 6x^2 + 6x + 8$$

## Q12

a)  $\sin 2x = \sqrt{2} \sin x \quad 0 \leq x \leq 2\pi$

$$2 \sin x \cos x - \sqrt{2} \sin x = 0$$

$$\sin x (2 \cos x - \sqrt{2}) = 0 \quad \checkmark$$

$$\sin x = 0 \text{ OR } \cos x = \frac{1}{\sqrt{2}}$$

$$x = 0, \pi, 2\pi \text{ OR } x = \frac{\pi}{4}, \frac{7\pi}{4} \quad \checkmark$$

b)  $\frac{dy}{dx} = 2xe^{-y}$  where  $y(0) = \ln 4$

No constant soln  $e^{-y} \neq 0$

Separate Variables  $\int e^y dy = \int 2x dx$

$$e^y = x^2 + C$$

$$y = \ln |x^2 + C| \quad \checkmark$$

$$y(0) = \ln 4 \quad \therefore \ln 4 = \ln |C|$$

$$C = 4$$

$$\therefore y = \ln |x^2 + 4| \quad \checkmark$$

c)  $P(4, -6) \quad Q(1, 2) \quad R(7, 5)$

$$\begin{aligned} \vec{QP} &= P - Q \\ &= \begin{pmatrix} 4-1 \\ -6-2 \end{pmatrix} \\ &= \begin{pmatrix} 3 \\ -8 \end{pmatrix} \\ &= 3\hat{i} - 8\hat{j} \end{aligned}$$

$$\begin{aligned} \vec{QR} &= R - Q \\ &= \begin{pmatrix} 7-1 \\ 5-2 \end{pmatrix} \\ &= \begin{pmatrix} 6 \\ 3 \end{pmatrix} \\ &= 6\hat{i} + 3\hat{j} \end{aligned}$$

Both Vectors either column vector or Cartesian

$$\begin{aligned} \vec{QP} \cdot \vec{QR} &= 3 \times 6 + (-8) \times 3 \\ &= 18 - 24 \\ &= -6 \\ &= |\vec{QP}| |\vec{QR}| \cos \angle PQR \end{aligned}$$

$$\begin{aligned} |\vec{QP}| &= \sqrt{3^2 + 8^2} = \sqrt{73} \\ |\vec{QR}| &= \sqrt{6^2 + 3^2} = \sqrt{45} = 3\sqrt{5} \end{aligned} \quad \checkmark$$

$$\therefore \cos \angle PQR = \frac{-6}{3\sqrt{5}\sqrt{73}}$$

$$= \frac{-2}{\sqrt{365}}$$

$$\begin{aligned} \angle PQR &= \cos^{-1} \frac{-2}{\sqrt{365}} \\ &\doteq 96^\circ \text{ (nearest degree)} \quad \checkmark \end{aligned}$$

d)  $1 + 3 + 9 + \dots + 3^{n-1} = \frac{1}{2}(3^n - 1)$

STEP A: Prove true for  $n=1$

$$\text{LHS} = 1 \quad \text{RHS} = \frac{1}{2}(3^1 - 1) = \frac{1}{2} \times 2 = 1 \quad \therefore \text{true for } n=1$$

STEP B: Assume true for  $n=k \quad k \in \mathbb{Z}^+$

$$\text{i.e. } 1 + 3 + 9 + \dots + 3^{k-1} = \frac{1}{2}(3^k - 1)$$

Prove true for  $n=k+1$

$$\text{i.e. } 1 + 3 + 9 + \dots + 3^{k-1} + 3^k = \frac{1}{2}(3^{k+1} - 1) \quad \checkmark \text{ Both statements}$$

$$\begin{aligned} \text{LHS} &= (1 + 3 + 9 + \dots + 3^{k-1}) + 3^k \\ &= \frac{1}{2}(3^k - 1) + 3^k \quad (\text{By assumption}) \\ &= \frac{3}{2} \times 3^k - \frac{1}{2} \\ &= \frac{1}{2}(3 \times 3^k - 1) \\ &= \frac{1}{2}(3^{k+1} - 1) \\ &= \text{RHS} \end{aligned}$$

$\therefore$  If true for  $n=k$  also true for  $n=k+1$

STEP C: By principle of Mathematical Induction conjecture true for all positive integer  $n$

Well structured argument

e)  $\cos^2 2x = \frac{1}{2}(1 + \cos 4x) \quad \checkmark$

$$\begin{aligned} \int_0^{\frac{\pi}{2}} \cos^2 2x \, dx &= \int_0^{\frac{\pi}{2}} \frac{1}{2} + \frac{1}{2} \cos 4x \, dx \\ &= \left[ \frac{x}{2} + \frac{1}{8} \sin 4x \right]_0^{\frac{\pi}{2}} \quad \checkmark \\ &= \left( \frac{\pi}{4} + \frac{1}{8} \sin \frac{2\pi}{2} \right) - 0 \\ &= \frac{\pi}{4} + \frac{0}{8} \\ &= \frac{4\pi + 0}{16} \\ &= \frac{\pi}{4} \end{aligned}$$

# QU 13

$$a) i) \sqrt{3} \cos \theta - \sin \theta = R \cos(\theta + \alpha)$$

$$R \cos \theta \cos \alpha - R \sin \theta \sin \alpha$$

$$\text{Equating coeffs } \cos \theta, R \cos \alpha = \sqrt{3} \quad (1)$$

$$R \sin \alpha = 1 \quad (2)$$

$$(1)^2 + (2)^2 \quad R^2(\cos^2 \alpha + \sin^2 \alpha) = 3 + 1$$

$$R^2 = 4$$

$$R = 2 \quad (R > 0)$$

$$\therefore \sin \alpha = \frac{1}{2} \quad \alpha = \frac{\pi}{6}$$

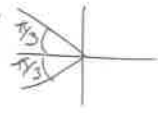
$$\text{ie } 2 \cos(\theta + \frac{\pi}{6})$$

$$(ii) 2 \cos(\theta + \frac{\pi}{6}) = -1 \quad 0 < \theta < 2\pi$$

$$\cos(\theta + \frac{\pi}{6}) = -\frac{1}{2} \quad \frac{\pi}{6} < \theta + \frac{\pi}{6} < \frac{13\pi}{6}$$

$$\theta + \frac{\pi}{6} = \frac{2\pi}{3}, \frac{4\pi}{3}$$

$$\theta = \frac{\pi}{2} \text{ OR } \frac{7\pi}{6}$$



$$b) T = A + Be^{-kt}$$

$$T = 22 + Be^{-kt} \quad (\text{room temp is } 22^\circ\text{C})$$

$$t=0 \quad T=4$$

$$4 = 22 + Be^0$$

$$B = (-18)$$

$$\therefore T = 22 - 18e^{-kt}$$

$$t=5 \quad T=12$$

$$12 = 22 - 18e^{-5k}$$

$$18e^{-5k} = 10$$

$$e^{-5k} = \frac{5}{9}$$

$$e^{5k} = \frac{9}{5}$$

$$5k = \ln(\frac{9}{5})$$

$$k = \frac{1}{5} \ln(\frac{9}{5})$$

$$t=8 \quad T=? \quad T = 22 - 18e^{-kt}$$

$$= 14.97^\circ\text{C}$$

Temp will be approx  $15^\circ\text{C}$

$$x(y) = \sqrt{a^2 - (x-a)^2} \quad (\text{NB: } a \text{ is a constant})$$

$$V_x = \pi \int y^2 dx$$

$$= \pi \int_0^b a^2 - (x-a)^2 dx \quad \checkmark \quad \text{"SHOW"}$$

$$= \pi \int_0^b a^2 - x^2 + 2ax - a^2 dx$$

$$= \pi \left[ -\frac{x^3}{3} + ax^2 \right]_0^b$$

$$= \pi \left[ \left( -\frac{b^3}{3} + ab^2 \right) - (0) \right] \quad \checkmark \quad \text{SHOW}$$

$$= \frac{\pi b^2}{3} (3a - b) \quad \text{u}^2$$

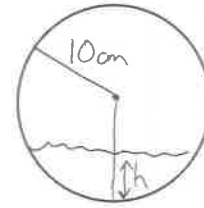
$$(ii) a=10 \quad \frac{dV}{dt} = 90 \text{ cm}^3/\text{min}$$

Let depth of water be  $h$ .

$$V = \frac{\pi h^2}{3} (30 - h)$$

$$= 10\pi h^2 - \frac{\pi h^3}{3}$$

$$\frac{dV}{dh} = 20\pi h - \pi h^2$$



$$\frac{dV}{dt} = \frac{dV}{dh} \times \frac{dh}{dt}$$

$$90 = (20\pi h - \pi h^2) \times \frac{dh}{dt} \quad \checkmark$$

$$\frac{dh}{dt} = \frac{90}{\pi h(20-h)}$$

$$\left( \frac{dh}{dt} \right)_{h=5} = \frac{90}{5\pi \times 15}$$

$$= \frac{6}{5\pi} \text{ cm/min} \quad \checkmark$$

$$d) \sin x - 7 \cos x = 5, \quad 0 \leq x \leq 2\pi$$

$$\text{Let } t = \tan \frac{x}{2} \therefore \sin x = \frac{2t}{1+t^2}, \quad \cos x = \frac{1-t^2}{1+t^2}$$

$$\frac{2t}{1+t^2} - \frac{7(1-t^2)}{1+t^2} = 5$$

$$2t + 7t^2 - 7 = 5 + 5t^2 \quad \checkmark$$

$$2t^2 + 2t - 12 = 0$$

$$t^2 + t - 6 = 0$$

$$(t+3)(t-2) = 0 \quad \checkmark$$

$$t = 2 \text{ OR } (-3)$$

$$\tan \frac{x}{2} = 2 \text{ OR } (-3)$$

$$x \div 1.107 \text{ OR } 1.892 \dots$$

$$x \div 2.21 \text{ OR } 3.79 \quad \checkmark \quad (2 \text{ dp})$$

# Q14

$$a) f(x) = \sec x \quad 0 \leq x < \frac{\pi}{2}$$

$$1 \leq f(x)$$

Inverse  $x = \sec y$   
 $x = \frac{1}{\cos y}$   
 $\cos y = \frac{1}{x}$  ✓ "SHOW"  
 $y = \cos^{-1}\left(\frac{1}{x}\right)$  ✓  
 $f(x) = \cos^{-1}\left(\frac{1}{x}\right)$

Range of  $f(x)$  is domain of  $f^{-1}(x)$   
 ie  $x \geq 1$

$$ii) \frac{d}{dx} f^{-1}(x) = \frac{d}{dx} \cos^{-1}\left(\frac{1}{x}\right)$$

$$= -\frac{1}{\sqrt{1-\left(\frac{1}{x}\right)^2}} \times \left(-x^{-2}\right)$$

$$= \frac{\frac{1}{x^2}}{\sqrt{1-\frac{1}{x^2}}} \quad (x \times x^2)$$

$$= \frac{1}{x\sqrt{x^2-1}} \quad (x \times x^2)$$

$$\text{as } x \rightarrow \infty \quad \frac{dy}{dx} \rightarrow 0^+ \quad \checkmark$$

Gradient decreases to zero  
 ie graph has horizontal asymptote ( $y = \frac{\pi}{2}$ )

$$b) i) P = \frac{2}{2-\sin t}$$

$$= 2(2-\sin t)^{-1}$$

$$\frac{dP}{dt} = -2(2-\sin t)^{-2} \times (-\cos t) \quad \checkmark$$

$$= \frac{2\cos t}{(2-\sin t)^2}$$

$$= \frac{4}{(2-\sin t)^2} \times \frac{\cos t}{2} \quad \checkmark \text{ "SHOW"}$$

$$= \frac{P^2 \cos t}{2} \quad (\text{as required})$$

$$ii) \frac{dP}{dt} = \frac{1}{2} P(2P-1) \cos t$$

Constant solns  $P=0$  and  $P=\frac{1}{2}$

$$\int \frac{1}{P(2P-1)} dP = \int \frac{\cos t}{2} dt$$

$$\int \frac{-1}{P} + \frac{2}{2P-1} dP = \frac{\sin t}{2} + C \quad \checkmark$$

$$\ln |2P-1| - \ln |P| = \frac{1}{2} \sin t + C$$

$$\ln \left| \frac{2P-1}{P} \right| = \frac{1}{2} \sin t + C$$

$$\left| \frac{2P-1}{P} \right| = e^{\frac{1}{2} \sin t + C}$$

$$= A e^{\frac{1}{2} \sin t} \quad (A > 0)$$

$$\frac{2P-1}{P} = A e^{\frac{1}{2} \sin t} \quad (A > 0 \text{ or } A < 0)$$

then include constant soln  $P=\frac{1}{2}$  ie  $A=0$

$$2P-1 = A e^{\frac{1}{2} \sin t}$$

$$P(2 - A e^{\frac{1}{2} \sin t}) = 1 \quad \checkmark$$

$$P = \frac{1}{2 - A e^{\frac{1}{2} \sin t}}$$

$$\text{but } t=0 \quad P=1 \quad \therefore 1 = \frac{1}{2 - A e^0} \Rightarrow A=1 \quad \checkmark$$

$$\therefore P = \frac{1}{2 - e^{\frac{1}{2} \sin t}}$$

$$(iii) P = \frac{2}{2-\sin t}$$

$$-1 \leq \sin t \leq 1$$

$$3 \geq 2-\sin t \geq 1$$

$$\therefore \frac{2}{3} \leq P \leq 2 \quad \checkmark$$

$$\text{ie } 0.667 \leq P \leq 2$$

Alternate Model

$$P = \frac{1}{2 - e^{\frac{1}{2} \sin t}}$$

$$-\frac{1}{2} \leq \frac{1}{2} \sin t \leq \frac{1}{2}$$

$$e^{-\frac{1}{2}} \leq e^{\frac{1}{2} \sin t} \leq e^{\frac{1}{2}}$$

$$2 - e^{\frac{1}{2}} \leq 2 - e^{\frac{1}{2} \sin t} \leq 2 - e^{-\frac{1}{2}}$$

$$\frac{1}{2 - e^{\frac{1}{2}}} \geq P \geq \frac{1}{2 - e^{-\frac{1}{2}}}$$

$$\frac{1}{2 - e^{\frac{1}{2}}} \leq P \leq \frac{1}{2 - e^{-\frac{1}{2}}} \quad \checkmark$$

$$0.718 \leq P \leq 2.847$$

c) If particles collide  $x_A = x_B$  at time  $T$

$$\therefore UT \cos \theta = R - \frac{UT \cos \theta}{2}$$

$$\frac{3UT \cos \theta}{2} = R$$

$$T = \frac{2R}{3U \cos \theta}$$

Also  $y_A = y_B$  at time  $T$

$$UT \sin \theta - \frac{1}{2} g T^2 = H + \frac{UT \sin \theta}{2} - \frac{1}{2} g T^2$$

$$H = \frac{UT \sin \theta}{2}$$

$$= \frac{U \sin \theta}{2} \times \frac{2R}{3U \cos \theta}$$

$$H = \frac{R \tan \theta}{3}$$

$$\text{but } \tan \alpha = \frac{H}{R}$$

$$\therefore \tan \alpha = \frac{1}{3} \tan \theta \text{ (as required)}$$